



Internet constellations

Four satellite constellations from SpaceX, Telesat, LeoSat and Kepler have been approved for low Earth orbit. <https://digit.in/dec18-73-1>



ISS completes 20 years!

The International Space Station completed 20 years in orbit this November 20. It was launched in 1998. <https://digit.in/dec18-73-2>

affordance of alternative activities for children in the community they are a part of and if other such external factors that also impact the duration children spend with a screen.

The results of the study do point to a correlation between increased screen time and lesser sleep – one that is quite insignificant when compared to other factors. For instance, the average sleep duration of a tech abstaining teen was found to be only a half hour more than that of one devoted to screens for about eight hours a day. There were a whole cluster of other reasons which caused a reduction in sleep times. Both regular bedtime discipline and earlier starts to school days were found to be way more significant in affecting sleep duration. The author has also indicated that future studies need to focus on data with a lower degree of imperfection to provide better insights into the real-world impact of screen time on the sleep cycles of non-adults. The study showed that there was actually very little correlation between the amount of screen time and the amount of sleep. The study also showed that the children of parents who enforced the guidelines prescribed for limiting screen time, did not sleep any better. You can read the full study here: digit.in/STStudy2.

PSYCHOLOGICAL AND PHYSICAL EFFECTS

While sleep is one of the first things that comes to mind when discussing the impact of screen time on non-adults, there are other areas impacted as well. The ubiquity of screens has made them significant influencers on the mental state of children and adolescents. Parents and caregivers are increasingly inclined to use videos and apps on smartphones and other smart devices to engage and educate non-adults. The impact of this has also been analysed by several studies, and the findings are more conclusive than those pertaining to sleep duration.

A study conducted earlier this year has found that increased screen time leads to lower mental well being

among 2 to 17 year-olds around us. While up to 1hr of screen time proved inconclusive, anything more than that led to adverse effects like “less curiosity, lower self-control, more distractibility, more difficulty making friends, less emotional stability, being more difficult to care for, and inability to finish tasks”, leading to a poorer overall state of mental health. Heavy users (more than 7 hours a day) are even identified to be twice as likely to be diagnosed with depression, anxiety and seek psychological help. This associa-



Shows like Sesame Street have been found to have a positive effect on the learning of children

tion was found to be more prominent in adolescents than younger children. Other studies in this area have reached similar results.

Things get interesting when we focus on younger children. Some studies indicate that there is a right amount of screen time that can actually have a positive effect on the cognitive abilities of children. In fact, there's enough evidence to establish that the factors affecting the mental health of children based on screen time are significantly more complex than those of sleep. For instance, productive content for children is usually designed keeping in mind the translation of skills from the two-dimensional format that screens can provide to the three-dimensional world around us, especially with the assistance of parents or immediate caregivers. While the well-structured material has been helpful, an analysis of most of the apps and programs

easily available online shows that these pay little attention to established and tested curriculums, or do not really encourage the learning process as much as seeming engaging to the child. In such cases, increases screen time definitely has the adverse effects mentioned earlier.

The impact is certainly not limited to the mind. Increased screen time often directly points to a sedentary lifestyle, which leads to conditions such as obesity among others. A study of 2-year-olds found that BMI increased for every hour per week of media consumed. Certain factors including food advertising and distraction during eating (which affects the perception of satiety cues) are also held responsible.

CONCLUSION

While the real world impact of screen time has been varying on

the non-adults that have been studied, there's no real confusion towards the fact that there is a negative impact, however significant or insignificant it might be. As many of these studies have shown, the need of the hour is to conduct detailed research involving cost-benefit analysis of screen time that goes down to the minutest level possible when it comes to accounting for variables in the study data. We're sure that we can pull that off, our future is at stake, after all! 📺

Sources/Suggested Reading:

- Oxford: <https://digit.in/screentime1>
- BBC: <https://digit.in/screentime2>
- ScienceDirect: <https://digit.in/screentime3>
- ScienceDirect: <https://digit.in/screentime4>
- PubMed: <https://digit.in/screentime5>
- Wiley: <https://digit.in/screentime6>
- AHA: <https://digit.in/screentime7>
- AAP: <https://digit.in/screentime8>
- Department of Health, Australia: <https://digit.in/screentime9>
- The Lancet: <https://digit.in/screentime10>